

FIELDCHECK · BUSINESS REQUIREMENTS DOCUMENT

Inspection and Field Data Capture Platform

Requirements, process models, multi-tenant deployment and supportability
— version 0.3

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1 Document control

Version	Date	Author	Summary of change
0.1	13 Sep 2026	Seshail Kamanna	First issue. Role catalogue, permission matrix, functional and non-functional requirements.
0.2	13 Sep 2026	Seshail Kamanna	Added BPMN process models, reference implementation evidence and revised open questions.
0.3	4 Oct 2026	Seshail Kamanna	Multi-tenancy requirements (section 6), supportability and operations (section 8), deployment options (section 10), updated implementation status, new acceptance criteria and open questions, legal notices and protective provisions (section 15). New visual identity. Diagram label collisions in the process models corrected.

1.1 Purpose of this revision

Version 0.2 described what the platform must do and how work moves between people. It assumed one organisation per deployment. Three things have changed since: the platform is now being prepared for more than one client, the commercial expectation is a managed service rather than a handed-over codebase, and the reference implementation has moved on — database migrations are in place and a single-file Windows evaluation build exists.

This revision therefore adds what a multi-client, managed service needs: tenant isolation, a layered approach to supportability, and the protective provisions that should accompany any document of this kind.

What changed and what did not

Requirement identifiers from versions 0.1 and 0.2 are unchanged and nothing has been renumbered, so review comments raised earlier still apply.

Section numbers have changed because new sections were added. Sections 6, 8, 10 and 15 are new. Sections 11, 12 and 13 are revised.

Open question OQ-11 (client-specific or reusable) remains closed as client-specific. Section 6 does not reopen it: multi-tenancy here means one operator serving several separate client organisations, not a shared template library.

1.2 How this document is organised

Section	Content
2–3	Scope, assumptions, roles and the permission matrix
4	Four BPMN process models
5	Functional requirements by role
6	Multi-tenancy requirements and tenancy models (new)
7	Non-functional requirements
8	Supportability and operations — the layered approach (new)
9–10	Integrations and deployment options (10 is new)

Section	Content
11	Reference implementation — evidence and honest gaps
12–14	Acceptance criteria, open questions, glossary
15	Legal notices and protective provisions (new)

2 Scope

The platform covers the full inspection cycle: authoring a checklist, scheduling it against a site or asset, executing it in the field with or without a network connection, reviewing and approving the result, and managing the corrective actions that arise from failures.

2.1 In scope

- Template authoring, versioning and controlled publication
- Scheduling, assignment, recurrence and adherence tracking
- Offline-capable execution on a mobile device, including photographic evidence, signatures and location capture
- Weighted scoring, critical-item handling and pass or fail determination
- Review, approval and return for correction
- Corrective and preventive action management through to verified closure
- Operational and management reporting, and data export
- Master data for sites, assets, users and roles
- Audit trail across every state change
- Operation as a managed service for more than one client organisation, with isolation between them (new in 0.3)

2.2 Out of scope for the first release

- Migration of historical paper inspection records
- Predictive analytics or machine-learning driven defect prediction
- Contractor onboarding and qualification management
- Asset maintenance planning beyond the corrective actions raised by an inspection
- A public self-service sign-up. Tenants are provisioned by the platform administrator

2.3 Assumptions

Ref	Assumption	If it proves wrong
AS-01	Data will be held in India and the solution must satisfy the Digital Personal Data Protection Act, 2023.	Hosting region and retention design change; no functional impact.
AS-02	Authentication will federate to the client's existing enterprise identity provider.	A local credential store and password policy must be built and operated.
AS-03	Offline capture is a hard requirement, not a convenience.	Materially simpler client, roughly a third less effort.

Ref	Assumption	If it proves wrong
AS-04	Historical paper records will not be migrated.	A digitisation workstream is needed, sized separately.
AS-05	The first clients will accept shared infrastructure with logical isolation, unless their contract says otherwise.	Those clients are deployed as dedicated stacks (the silo model in section 6.2). Cost per client rises; code does not change.
AS-06	The operator, not the client, runs the platform day to day.	Client-run deployments need the runbooks and support console of section 8 handed over with training.

3 Roles

Seventeen roles are defined. Sixteen were identified in version 0.1 and are unchanged because the process models in section 4 refer to them directly. R-17 is new and exists only because the platform is now operated for several clients.

Ref	Role	Primary responsibility
R-01	Super administrator	Tenant configuration, security policy, data retention.
R-02	Administrator	Master data, users, roles and site structure.
R-03	Template author	Builds and maintains inspection templates.
R-04	Template approver	Confirms a template reflects the control requirement before publication.
R-05	Scheduler	Creates the inspection programme and assigns work.
R-06	Field inspector	Executes inspections and captures evidence.
R-07	Reviewer / approver	Accepts, returns or rejects a submitted inspection.
R-08	Action owner	Carries out a corrective action and records completion.
R-09	Action verifier	Confirms an action was effective and closes it.
R-10	Site manager	Accountable for inspection outcomes at one site.
R-11	Quality / EHS manager	Owens the programme across sites; sets pass marks and criticality.
R-12	Executive	Reads dashboards and trends; no transactional rights.
R-13	Internal auditor	Read-only access across all records and the audit trail.
R-14	External contributor	Contractor or third party executing a limited set of templates.
R-15	Data analyst	Exports and analyses; no write access to operational records.
R-16	Support / help desk	Assists users; cannot alter inspection content or outcomes.
R-17	Platform administrator (new)	Provider-side. Creates and suspends tenants, sets quotas, monitors service health. No routine access to any tenant's inspection content (see MT-07).

3.1 Role and permission matrix

C create, R read, U update, D delete, A approve. A blank cell means no access. Site scoping applies on top of this matrix: a user sees only the sites they are assigned to, except where marked as cross-site. Every role is also confined to its own tenant (section 6).

Role	Templates	Schedule	Inspections	Actions	Master data	Reports
Super administrator	CRUD	CRUD	R	R	CRUD	R
Administrator	R	CRU	R	R	CRUD	R
Template author	CRU		R		R	R
Template approver	RA		R		R	R
Scheduler	R	CRUD	R	R	R	R
Field inspector	R	R	CRU own	CRU own	R	
Reviewer / approver	R	R	RA	CRU	R	R
Action owner			R	RU own	R	
Action verifier			R	RA	R	R
Site manager	R	RU	R	CRUA	R	R
Quality / EHS manager	RU	CRUD	RA	CRUA	R	R cross-site
Executive			R	R		R cross-site
Internal auditor	R	R	R	R	R	R cross-site
External contributor	R limited	R own	CRU own	R own		
Data analyst	R	R	R	R	R	R cross-site
Support / help desk	R	R	R	R	R	
Platform administrator	—	—	—	—	—	—

The Platform administrator row is deliberately empty. That role manages tenants, not tenants' content. Where support genuinely needs to see content, section 6 requires a time-limited, justified, audited and notified break-glass procedure rather than a standing permission.

The matrix is a build artefact, not documentation

In the reference implementation this table is a single dictionary in the authorisation module rather than conditional checks spread through the code. That means the access model can be read in one screen and tested directly — and it is, by the automated test suite.

4 Process models

Four end-to-end processes describe how work moves through the platform. They are drawn in BPMN 2.0 notation: a pool holds the process, lanes hold the participants, rounded boxes are tasks, diamonds are exclusive decisions, thin circles start a flow and thick circles end it. A dashed line is a message or data flow rather than a sequence flow.

Where a control is enforced by the system rather than by procedure, that is summarised in the table after the diagrams, because it is the difference between a rule people are asked to follow and a rule they cannot break. The processes are unchanged from version 0.2 and are the same in every tenant; tenant differences are limited to configuration (MT-05).

Model	Process	Lanes
P-01	Template authoring, review and publication	Template author, Quality manager, Platform
P-02	Inspection scheduling and execution	Scheduler, Field inspector, Device and platform
P-03	Review, approval and record closure	Field inspector, Reviewer, Platform
P-04	Corrective and preventive action lifecycle	Platform and raiser, Action owner, Verifier

The diagrams follow on landscape pages so the text in them is readable.

P-01 Template authoring, review and publication

A template is the control instrument. Getting its governance right matters more than authoring convenience, because every inspection result downstream inherits whatever the template says.

P-01 Template authoring, review and publication

From a control requirement being identified to a template version reaching inspectors

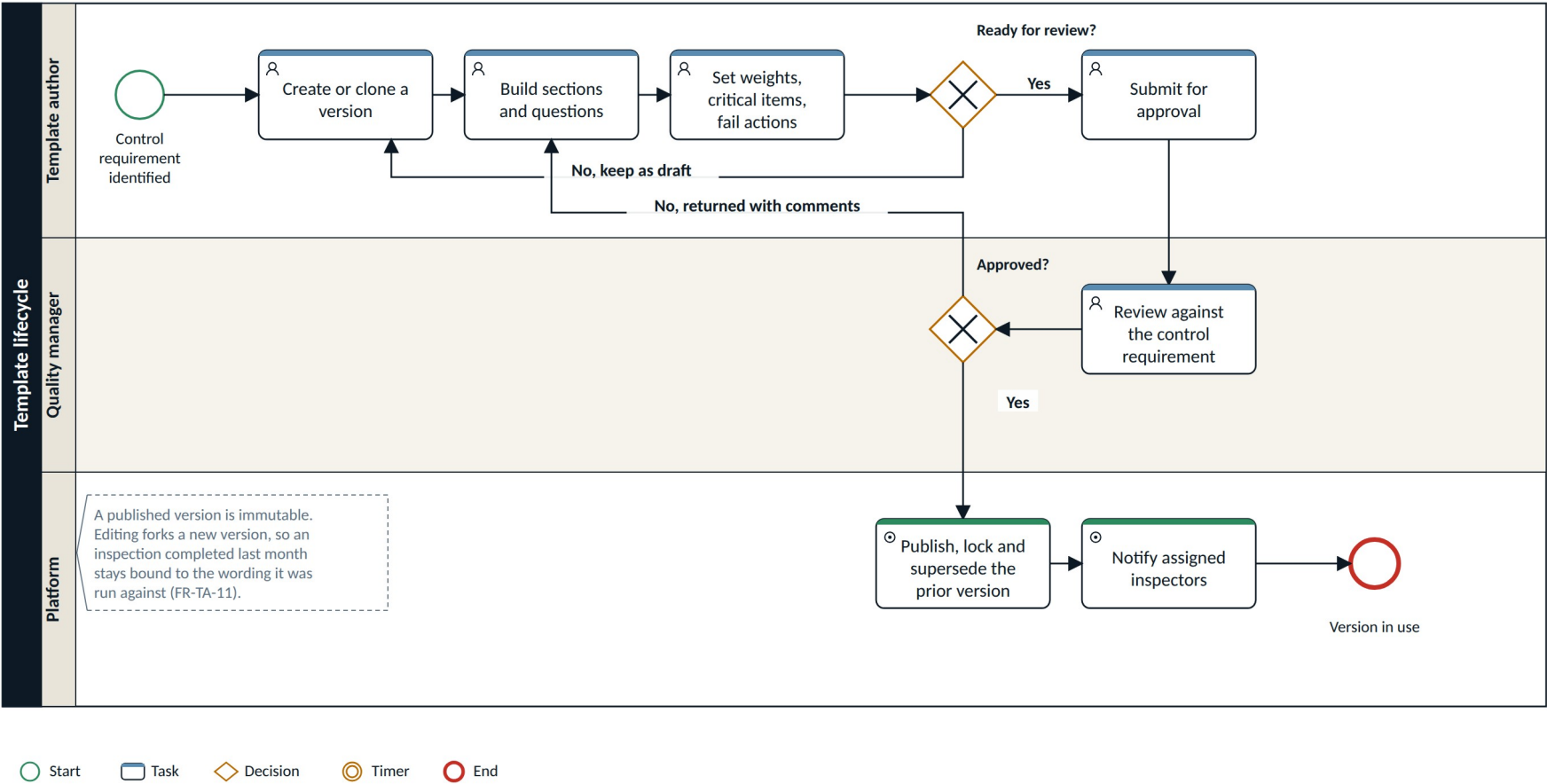


Figure P-01 — Template authoring, review and publication

P-02 Inspection scheduling and execution

The offline branch determines whether the platform is usable. On a plant floor or in a warehouse aisle, no signal is the normal condition rather than an edge case.

P-02 Inspection scheduling and execution

The offline path is the normal case on a plant floor, not the exception

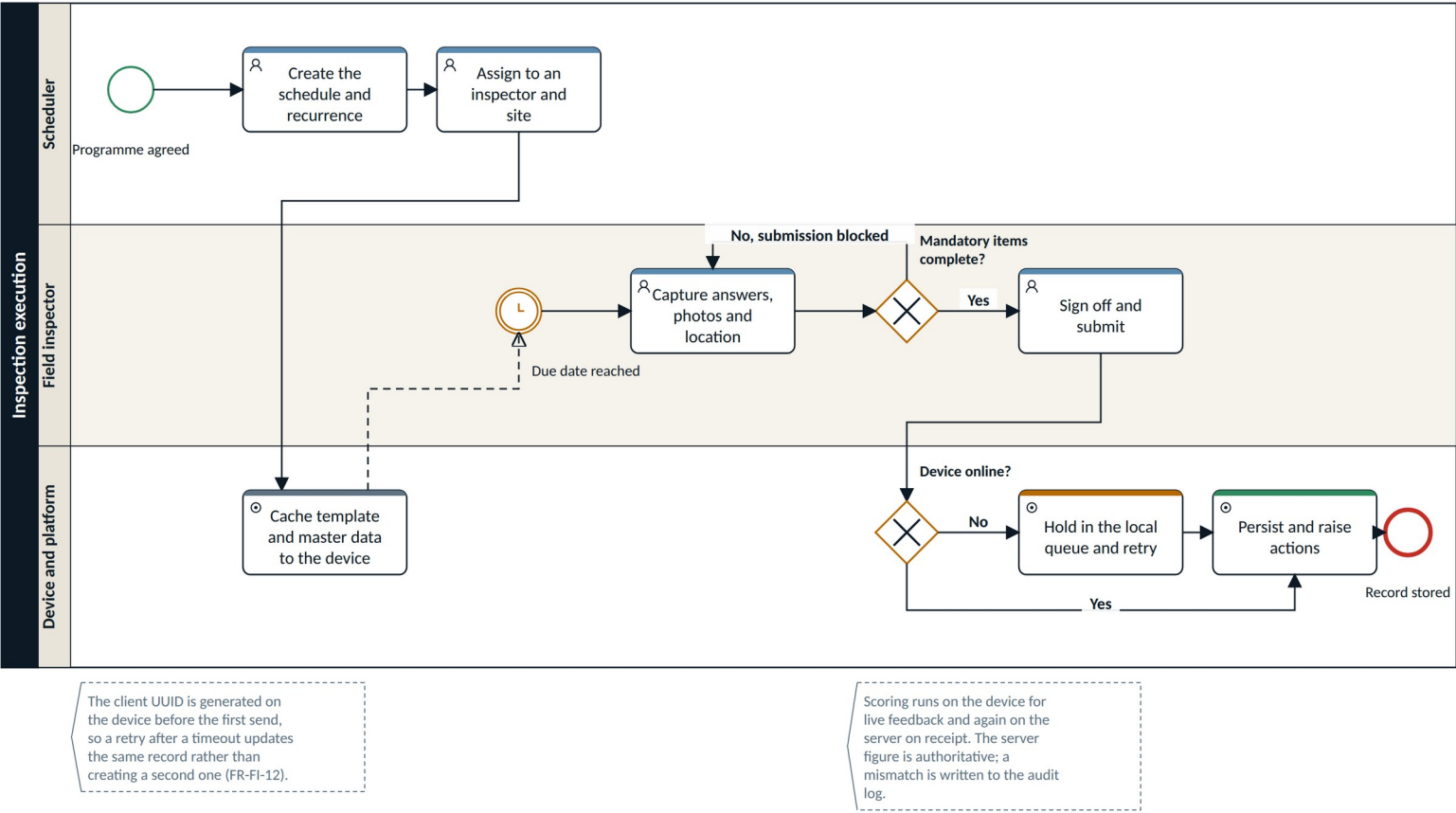


Figure P-02 — Inspection scheduling and execution

P-03 Review, approval and record closure

Review is where the record acquires its evidential weight. The control that matters is that the person who did the work is not the person who signs it off.

P-03 Review, approval and record closure

Segregation of duties is enforced by the platform, not left to procedure

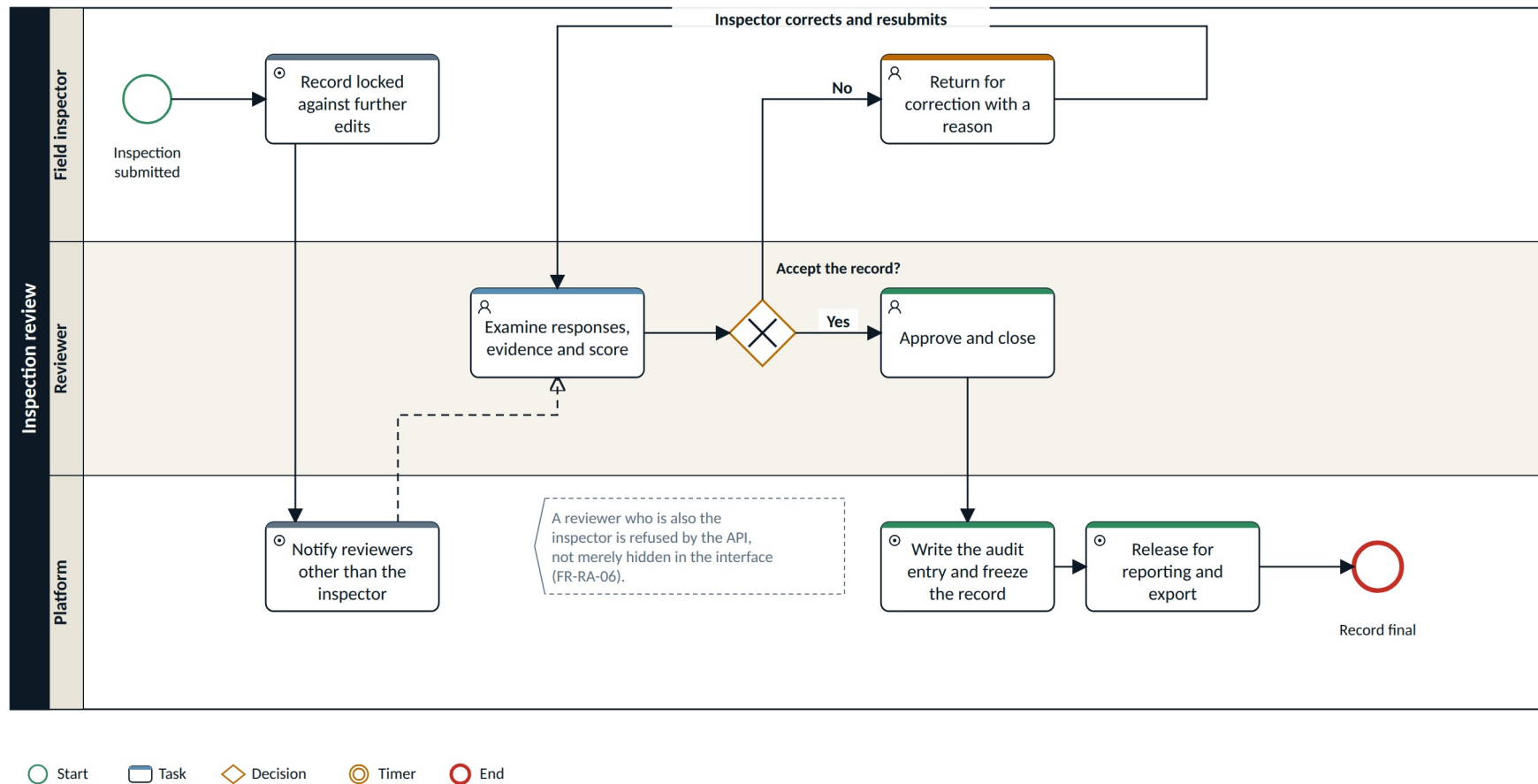


Figure P-03 — Review, approval and record closure

P-04 Corrective and preventive action lifecycle

Actions turn an inspection from a record into an improvement. Most of the value of the platform, and most of the argument about it, lives in this process.

P-04 Corrective and preventive action lifecycle

Raised automatically from a failed item, or manually by any authorised user

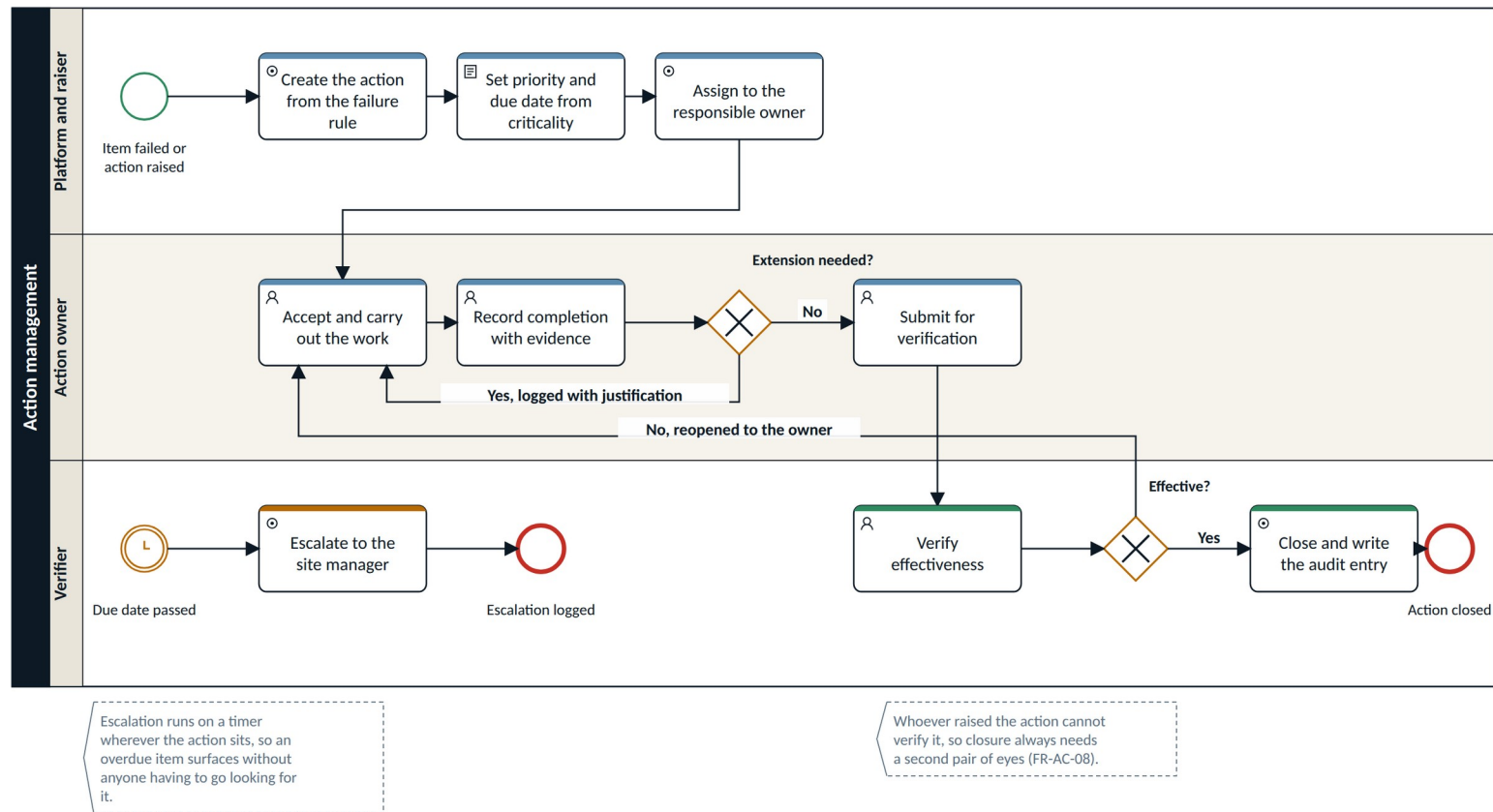


Figure P-04 — Corrective and preventive action lifecycle

Controls the platform enforces

These are the points in the four processes where the platform, not procedure, decides. Each is covered by an automated test or is a stated requirement.

Process	What the platform enforces
P-01	A draft may cycle between author and approver as many times as needed. Nothing is visible to inspectors until it is published. Publication is irreversible. A published version cannot be edited; editing forks a new version and supersedes the old one. Inspections stay bound to the template version they were executed against, so a record completed six months ago still reads against the wording in force at the time (FR-TA-11).
P-02	Template and master data are cached to the device at assignment, so the inspector is never blocked waiting for a download at the point of use. Submission is refused while any mandatory item is outstanding. The interface lists what is missing rather than simply disabling the button (FR-FI-14). A client-generated identifier travels with the record. A retry after a timeout updates the same record instead of creating a duplicate — the failure mode that most often corrupts field data (FR-FI-12). Scoring runs twice: on the device for live feedback, and again on the server on receipt. The server result is authoritative.
P-03	On submission the record locks. The inspector cannot alter answers or evidence afterwards. An inspector cannot review their own inspection. This is refused by the application programming interface, not merely hidden in the interface, so it cannot be bypassed by a crafted request (FR-RA-06). Returning a record for correction requires a reason, which is written to the audit trail and shown to the inspector. Once approved the record is frozen and released to reporting and export.
P-04	A failed item raises its action automatically, using the wording and due-date rule set on the question when the template was authored. Critical items get a twenty-four hour due date; others get seven days. A due-date extension is permitted but requires a justification, and both the old and new dates stay in the record. Whoever raised an action cannot verify and close it. Closure always requires a second person with verification rights (FR-AC-08). An action found not to be effective reopens to the owner rather than closing with a note, so the register never shows closed work that did not work. Escalation runs on a timer wherever the action sits in the flow, so an overdue item surfaces without anyone having to go looking for it.

5 Functional requirements

Requirements are grouped by the role that exercises them and prioritised using MoSCoW. Identifiers are unchanged from version 0.1. Only the requirements referenced by the process models are reproduced in full; the complete catalogue is carried forward from version 0.1 without amendment.

5.1 Template author (FR-TA)

Ref	Requirement	Priority
FR-TA-01	Create a template containing ordered sections, each holding ordered questions.	Must
FR-TA-02	Support answer types: pass/fail/not applicable, single select, multi select, numeric reading, free text, date, photograph, signature, asset lookup and barcode scan.	Must
FR-TA-03	Set a weight on each scored question.	Must
FR-TA-04	Set a pass mark for the template as a whole.	Must
FR-TA-05	Show or hide a question conditionally on the answer to an earlier question.	Should
FR-TA-08	Mark a question as critical, so that failing it fails the inspection regardless of the numeric score.	Must
FR-TA-09	Require a photograph when an item is failed.	Should
FR-TA-10	Define the corrective action to be raised when an item is failed, including its wording and due-date rule.	Must
FR-TA-11	Version templates. A published version is immutable; editing creates a new version and supersedes the prior one.	Must

5.2 Field inspector (FR-FI)

Ref	Requirement	Priority
FR-FI-01	Execute an assigned inspection against a site and, where relevant, a specific asset.	Must
FR-FI-05	Attach photographic evidence to any item, captured through the device camera.	Must
FR-FI-06	Capture a signature at sign-off.	Must
FR-FI-07	Capture location coordinates at the point of execution.	Must
FR-FI-10	Execute a complete inspection with no network connection available.	Must
FR-FI-12	Synchronise on reconnection without creating duplicate records, using a client-generated identifier.	Must
FR-FI-14	Be prevented from submitting while a mandatory item is outstanding, and be shown which items remain.	Must
FR-FI-15	Supply a mandatory reason when marking an item not applicable.	Should

FR-FI-15 is a control, not a nicety

Items marked not applicable are excluded from both the numerator and the denominator of the score. That is the defensible treatment, but it means an inspector can raise a score by marking difficult items not applicable. The mandatory reason is the only thing standing between the scoring model and quiet inflation. It should not be dropped to save effort.

5.3 Reviewer and approver (FR-RA)

Ref	Requirement	Priority
FR-RA-01	See all inspections submitted within the reviewer's site scope.	Must

Ref	Requirement	Priority
FR-RA-02	Examine every response, attachment and score before deciding.	Must
FR-RA-03	Approve an inspection, which freezes the record.	Must
FR-RA-04	Return an inspection for correction, with a mandatory reason.	Must
FR-RA-06	Be prevented from reviewing an inspection the reviewer personally executed.	Must

5.4 Corrective action (FR-AC)

Ref	Requirement	Priority
FR-AC-01	Raise an action automatically when an item is failed, using the rule defined on the question.	Must
FR-AC-02	Raise an action manually, independent of any inspection.	Must
FR-AC-03	Assign and reassign an owner, recording who reassigned and when.	Must
FR-AC-04	Record completion with supporting evidence.	Must
FR-AC-05	Extend a due date, with a mandatory justification and retention of the original date.	Should
FR-AC-06	Escalate an overdue action to the site manager automatically.	Must
FR-AC-07	Reopen an action found not to be effective.	Must
FR-AC-08	Prevent the person who raised an action from verifying and closing it.	Must

6 Multi-tenancy requirements

A tenant is one client organisation using the platform: its users, sites, templates, inspections, actions, evidence and audit trail. Multi-tenancy means one operator serves several tenants, and no tenant can see, change or infer anything about another.

Where the reference implementation stands

The reference implementation is single-tenant. It has no tenant identifier on any record, and one deployment serves one organisation. Everything in this section is a requirement, not a description of what exists. Section 11.2 sizes the gap.

6.1 Why isolation is the hard part

Inspection records are evidence. A cross-tenant leak is not a cosmetic defect: it exposes one client's safety failures, photographs, locations and the names of their people to another client. The requirement is therefore stated as a property that must hold under error, not only under correct use. If a developer forgets a filter, the platform must still refuse.

6.2 Tenancy models considered

Model	How it works	Isolation	Cost per tenant	Fits
Silo	A full stack per client: own application instances, own database, own sub-domain.	Strongest. Separate failure and compliance boundary.	Highest	Regulated or large clients; contractual data separation; data-residency demands.
Schema or database per tenant	One application fleet and one database server; a schema or database per client.	Strong. Cross-tenant query is structurally harder.	Medium	10–200 mid-size clients. Migrations must run once per tenant, which becomes painful at scale.
Pooled	One application, one database; every row carries a tenant identifier.	Weakest in principle; enforced in software and by the database.	Lowest	Many small clients, subscription pricing.

Recommendation: pooled by default, silo on demand

Build the pooled model as the standard, with database row-level security as a second lock. Keep the silo option open from the start, because the same release deployed with its own database is the answer for any client whose contract demands it (MT-12).

This is a recommendation, not a decision. It is carried as open question OQ-12 and depends on what the first clients will sign.

6.3 Requirements

Ref	Requirement	Priority
MT-01	Every business record belongs to exactly one tenant, identified by a tenant identifier. This includes users, sites, assets, templates, inspections, answers, evidence files, actions, schedules and audit entries.	Must
MT-02	The tenant is derived from the authenticated session. It is never accepted from a request parameter, header or body supplied by the client.	Must
MT-03	Tenant filtering is applied in one central place, and is backed by database row-level security so that a missing filter in application code returns nothing rather than another tenant's rows.	Must
MT-04	Photographs, signatures and other evidence are stored in a per-tenant namespace. Retrieval checks the requester's tenant, not only the file name.	Must
MT-05	Each tenant has its own configuration: name, branding, retention period, default pass mark and sign-in method.	Should
MT-06	Email addresses are unique within a tenant. The rule for a person who works for two tenants is explicit and tested.	Should
MT-07	The Platform administrator can create, suspend and delete tenants and see usage, but cannot read tenant content in the ordinary course. Break-glass access is time-limited, requires a written justification, is recorded in the tenant's own audit trail and is notified to the tenant administrator.	Must
MT-08	An automated test signs in as one tenant and attempts to read, change and delete another tenant's records at every endpoint, expecting refusal. The test is part of every release.	Must

Ref	Requirement	Priority
MT-09	A tenant can obtain a complete, machine-readable extract of its data, including evidence files and the audit trail.	Must
MT-10	Tenant offboarding deletes all tenant data after the agreed retention period, and issues written confirmation that deletion, including from backups on their normal expiry, has been completed.	Must
MT-11	Quotas apply per tenant — users, storage and request rate — so one tenant cannot degrade the service for the others.	Should
MT-12	The same release can be deployed as a dedicated stack with its own database for a tenant that requires it, with no code change.	Must
MT-13	The offline synchronisation endpoint rejects any record whose tenant does not match the signed-in user's tenant, and client-generated identifiers cannot collide across tenants.	Must

6.4 Target deployment shape

For the pooled standard, the production shape is a stateless application tier of at least two instances behind a load balancer, a managed PostgreSQL service with automated backups and point-in-time recovery, object storage for evidence with a per-tenant prefix, and a wildcard DNS name and certificate so each tenant has its own address. A cache is optional until session or rate-limit volume justifies one.

- **Tenant addressing.** Each tenant has its own sub-domain. Custom client domains with automatic certificates follow later.
- **Hosting.** On Azure: Container Apps or App Service, Azure Database for PostgreSQL Flexible Server and Blob Storage. On AWS: ECS Fargate, RDS and S3. Either needs a managed PostgreSQL; SQLite is not a production option beyond a single machine.
- **Data residency.** Which region each tenant's data sits in is a per-tenant decision recorded at provisioning (OQ-13).
- **Offline clients.** Signed-in devices cache only their own tenant's templates and master data. Signing out clears them.

7 Non-functional requirements

Ref	Area	Requirement
NFR-01	Performance	A template of up to 200 questions opens on a mid-range device within three seconds.
NFR-02	Performance	A completed inspection with twenty photographs synchronises within sixty seconds on a 3G connection.
NFR-03	Availability	99.5 per cent during Indian business hours, measured monthly, in the first year; target 99.9 per cent thereafter (see section 8.3).
NFR-04	Offline	A device holds at least thirty completed inspections and their evidence pending synchronisation.
NFR-05	Security	Credentials are stored using a modern adaptive hash. Sessions expire and are revocable.
NFR-06	Security	Every state change writes an audit entry carrying actor, action, entity, timestamp, tenant and source address.

Ref	Area	Requirement
NFR-07	Data protection	Personal data is handled under the Digital Personal Data Protection Act, 2023. Retention is configurable per tenant and enforced.
NFR-08	Usability	The execution screen is operable one-handed on a phone, with touch targets no smaller than 48 pixels.
NFR-09	Usability	Colour is never the sole carrier of meaning. Contrast meets WCAG 2.1 AA.
NFR-10	Maintainability	The authorisation model is expressed in one place and covered by automated tests.
NFR-11	Isolation	No request, query, file retrieval, export or background job can return another tenant's data, including under application error (MT-03, MT-08).
NFR-12	Recoverability	Recovery point objective of five minutes or better and recovery time objective of four hours, through point-in-time recovery. A restore is rehearsed monthly and the result recorded.
NFR-13	Observability	Every request carries an identifier that appears in the response shown to the user, in the logs and in client error reports, so a single quoted code locates the whole story.
NFR-14	Change safety	Every release passes automated tests and a staging environment, can be rolled back in one step, and carries only database migrations that remain compatible with the previous release.
NFR-15	Portability	The platform runs from a container image on any host that runs containers, and from a single-file Windows build for evaluation. No component is specific to one cloud.

8 Supportability and operations

The aim is that any problem is detected automatically, located to one layer within minutes, fixed by a written procedure, and learned from. This is a requirement on the platform and on the service wrapped around it. The approach is layered because a fault is easiest to find when the first question is *where*, not *what*.

Status

Most of this section is a requirement, not a description of the reference implementation. The right-hand column of the table in 8.1 states what exists today.

8.1 Seven layers

#	Layer	Question it answers	What is required	Today
1	Experience	What did the user actually see?	Client error reporting with release, device, tenant and the user's last actions. Visible offline-queue status.	Offline status shown. No error reporting.
2	Edge	Can the user reach us at all?	External uptime probes, certificate-expiry alerts, rate limiting.	Not built.
3	Application	Is the software behaving correctly?	Structured logs with request, tenant and user identifiers. Separate liveness and readiness checks. Request, error and latency metrics.	Basic health endpoint. Generic error message.

#	Layer	Question it answers	What is required	Today
4	Data	Is the data safe and consistent?	Point-in-time recovery, monthly restore drill, slow-query and capacity alerts, schema changes only through migrations.	Migrations in place; application refuses to start on a mismatched schema. No backups.
5	Business integrity	Is the meaning right, even if nothing crashed?	Scheduled checks: inspections stuck in submitted, failed items without an action, rejected syncs, evidence referenced but missing.	Not built.
6	Security and audit	Who did what, and is anything abnormal?	Audit trail extended to administrative and support actions. Alerts on repeated sign-in failure, role change and cross-tenant attempts. Dependency scanning.	Audit trail on inspection and action state changes.
7	Operations and governance	Who responds, how fast, and what changes?	On-call rota, severity levels, runbooks, release and rollback procedure, monthly incident review.	Not built.

Layers 5 and 7 are what make a platform managed rather than merely monitored. Layer 5 catches the failures technical monitoring never sees, such as an inspection that submitted correctly and then sat unreviewed for a month.

8.2 Operational requirements

Ref	Requirement	Priority
OPS-01	Every error shown to a user carries a reference that identifies the request, tenant and time. Quoting it to support locates the failure in the logs without further questions.	Must
OPS-02	A support console, available to the Support role and the Platform administrator, allows a user, tenant, inspection or action to be looked up, and shows its synchronisation state, recent errors and audit history.	Must
OPS-03	The support console can retry a stuck synchronisation, revoke a session and reset a credential. Each action is audited, attributable and visible to the tenant administrator.	Must
OPS-04	Alerts fire on what a user would feel: error rate, failed sign-ins, failed synchronisation, missed backups. Everything else goes into a daily digest. Every alert links to its runbook.	Must
OPS-05	A written runbook exists for each known failure — confirmation of cause, fix, escalation, verification of recovery. A runbook is added or amended after every severity-1 or repeated incident.	Must
OPS-06	A blameless post-incident review follows every severity-1 incident and any incident that recurs, ending in a change to a runbook, an alert or a test.	Should
OPS-07	Risky changes can be switched off without redeploying, using feature flags.	Should
OPS-08	A status page shows current and past incidents, so tenants are told before they have to ask.	Should
OPS-09	Failed operations that are safe to retry are retried automatically with back-off. Submissions are idempotent, so a retry cannot duplicate a record (FR-FI-12).	Must

8.3 Service targets (proposed)

These are proposals for the service schedule and are not yet agreed with any client.

Severity	Meaning	Acknowledge	Updates
1	Platform unavailable, or data at risk	15 minutes, around the clock	Every 30 minutes until resolved
2	A feature broken for many users, or a tenant degraded	1 hour in business hours	Every 2 hours
3	A problem affecting one user or with a workaround	1 business day	On resolution
4	Question or minor defect	5 business days	On resolution

Target	First year	Thereafter
Availability, measured monthly	99.5%	99.9%
Recovery point (data lost at worst)	5 minutes	5 minutes
Recovery time (back in service)	4 hours	1 hour
Restore drill	Monthly	Monthly

9 Integrations

Ref	Integration	Priority
INT-01	Enterprise identity provider for single sign-on (SAML 2.0 or OpenID Connect), configured per tenant.	Must
INT-02	Email and push notification delivery.	Must
INT-03	WhatsApp or SMS for inspectors without a corporate mailbox.	Could
INT-04	Enterprise resource planning system for asset master data.	Should
INT-05	Enterprise asset management system for work order creation from actions.	Should
INT-06	Business intelligence layer for ad-hoc analysis.	Should

10 Deployment options

The same codebase is packaged several ways. They are not interchangeable, and the right one depends on what the deployment is for.

Option	What it is	Use for	Limits
Windows evaluation build	A single FieldCheck.exe with built-in database. Opens in the browser on the same machine. With - -lan other devices on the same network can connect.	Demonstration, a client meeting, a pilot with a handful of users.	SQLite, one machine, no backups, no HTTPS, demonstration credentials. Not for real data.
Single-tenant container	Docker Compose or a single image with PostgreSQL.	A pilot or a dedicated client stack (the silo model).	One organisation per deployment. Operator carries backups and monitoring.

Option	What it is	Use for	Limits
Free-tier test hosting	Small virtual machine or platform service with automatic HTTPS.	Testing with real network conditions during development.	Small capacity, no availability commitment.
Multi-tenant production	Section 6.4: stateless application tier, managed PostgreSQL, object storage, wildcard addressing.	A managed service for several clients.	Requires section 6 to be built and section 8 to be operating.

Evaluation build credentials

The demonstration accounts and their shared password are published in the user manual. They exist so a demonstration works out of the box. An instance holding any real data must be started with demonstration data disabled and with credentials issued individually.

11 Reference implementation

A working implementation was built alongside this document to test whether the requirements hold together in practice. It is a complete application with an automated test suite, and it is the source of the screenshots here and in the user manual. Its purpose in this document is evidential: a requirement that has been built and tested has been shown to be implementable.

11.1 What the implementation demonstrates

Requirement	Demonstrated by
FR-TA-11 template immutability	Publishing locks the version; editing forks a new one. Covered by an automated test.
FR-FI-12 duplicate-free sync	Resubmitting with the same client identifier returns the original record. Covered by an automated test.
FR-FI-14 submission gate	Submission is refused with the outstanding items listed. Covered by an automated test.
FR-RA-06 review segregation	An inspector reviewing their own work is refused by the interface. Covered by an automated test.
FR-AC-01 automatic actions	A failed critical item raises its action with a twenty-four hour due date on submission.
FR-AC-08 closure segregation	The raiser attempting to close their own action is refused. Covered by an automated test.
NFR-08, NFR-09 field usability	Execution screen verified at 390 pixels wide with 48 pixel targets and AA contrast.
Schema control	Database changes are made only through versioned migrations. The application refuses to start against a database at the wrong revision, and the tests build their schema from the migrations.
NFR-15 portability	Runs from a container image, and from a single-file Windows build that has been built and exercised end to end.

The backend test suite currently runs 23 tests. All pass on Python 3.14.8 with the library versions pinned for the desktop build.

11.2 What the implementation does not yet cover

These gaps are stated plainly because they affect estimates and should not be discovered during a demonstration.

Gap	Requirement	Indicative size
Multi-tenancy — no tenant identifier, central filtering, row-level security, per-tenant evidence storage or isolation tests	MT-01 to MT-13	Largest item. Roughly two to three weeks for the pooled core, and a further two to three for onboarding, branding and platform administration screens.
Supportability — no correlation identifiers, structured logs, alerts, backups, support console or runbooks	NFR-12 to NFR-14, OPS-01 to OPS-09	Roughly six to eight weeks for one engineer, overlapping the multi-tenant work.
Single sign-on. Authentication is local; federation is designed for but not built.	INT-01	Per identity provider.
Notification delivery. Notifications exist as records; nothing dispatches email, push or WhatsApp. Needs a message broker and a worker process.	INT-02, INT-03	Moderate.
Geofence enforcement. Coordinates are captured but a submission taken away from the asset is not rejected.	OQ-07	Small, once the rule is decided.
Mandatory reason for a not-applicable answer is not implemented.	FR-FI-15	Small.
Enterprise resource planning and asset management integration.	INT-04, INT-05	Per system.

Estimates are indicative and carry no commitment (section 15.4). Database schema migrations, listed as a gap in version 0.2, are now complete.

11.3 Interface evidence

The screens below are taken from the running implementation in the current visual identity. The full walkthrough, including the mobile layouts, is in the accompanying user manual. The photograph shown is a synthetic stand-in created for the demonstration.

FieldCheck

Inspection & data capture

Seshail Kamanna · seshail.com

My work

Inspections

Actions

Schedule

Templates

Connected

FieldCheck

SiteAll sites

RK

Rohan Kulkarni

Inspector

Sign out

Forklift pre-start check

Chakan Plant · Forklift 3T — Yard · v1

0%1/6

ScoreAnswered

Critical item failed

Inspection context

Site

Chakan Plant

Asset or location

FLT-007 · Forklift 3T — Yard

Capture location

Location not captured

01 Visual check

* No visible hydraulic leaks

Critical

Pass

Fail

N.A.

Add photo

Edit note

Note: Hydraulic hose weeping at the left mast cylinder.

Action on submission: Withdraw the forklift from service — hydraulic leak

Assigned to the site manager, due in 24 hours.

* Tyres free of damage and correctly inflated

Pass

Fail

N.A.

Add photo

Add note

Figure 11.1 — A failed critical item during execution. The hazard marking and the action to be raised are both visible before the inspector submits.

FieldCheck

Inspection & data capture

Seshail Kamanna · seshail.com

Dashboard

Inspections

Actions

Schedule

Templates

Connected

Dashboard

SiteAll sites

AD

Anil Deshmukh

manager

Sign out

87%

Average inspection score

5 in the last 90 days

100%

Schedule adherence

0 past due

6

Open actions

3 overdue

100%

Closed on time

Against the agreed due date

2

Critical failures

Inspections failing a critical item

3

Awaiting review

Submitted, not yet approved

Score trend

1.00

75

50

Most frequently failed items

Tyres free of damage and correctly inflated

1x100%

Pressure gauge reads within the green band

1x100%

No visible hydraulic leaks

1x100%

Open action ageing

0-7 days

6100%

8-30 days

00%

31-90 days

00%

90+ days

00%

By site

Site

Inspections

Average score

Chakan Plant

4

89.3%

Bhiwandi Warehouse

1

77.8%

Export

Inspections (CSV)

Actions (CSV)

Exports respect your site access. Downloads open with your current session.

Figure 11.2 — Management dashboard, showing schedule adherence, open action ageing and the most frequently failed items.

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12 Acceptance criteria

Ref	Criterion
AC-01	An author can build, version and publish a template with conditional logic and critical items, without developer assistance.
AC-02	An inspector can complete an inspection with photographs and a signature end to end with the device in flight mode, and synchronise on reconnection with no duplicate created.
AC-03	Submission is refused while a mandatory item is outstanding, and the outstanding items are named.
AC-04	A failed critical item fails the inspection irrespective of the numeric score, and raises its action automatically.
AC-05	An inspector cannot review their own inspection and the raiser of an action cannot close it, in both cases enforced server-side.
AC-06	An overdue action escalates to the site manager without manual intervention.
AC-07	Every state change is recoverable from the audit trail, including who made it and when.
AC-08	The execution screen is usable one-handed on a 5-inch device in daylight, confirmed by inspector testing on site.
AC-09	With two tenants loaded, a user of the first cannot read, change, delete, export or download evidence belonging to the second, at any endpoint, including by guessing identifiers. The isolation test of MT-08 passes on every release.
AC-10	A backup is restored into a clean environment to a chosen point in time, the result is checked, and the exercise is recorded. This is repeated monthly.
AC-11	Given only the reference code shown on an error screen, a support engineer finds the request, the tenant and the cause in the logs within five minutes.
AC-12	A tenant extract and a verified tenant deletion are carried out on a test tenant, and the written confirmation is produced.

13 Open questions

Revised from version 0.2. Questions answered by the reference implementation, or since decided, are closed and marked as such. Questions OQ-12 to OQ-15 are new.

OQ-11 remains settled: this is a client-specific solution rather than a reusable practice asset. The template library should be built from the client's own control set rather than generic industry checklists, and the master data model, site hierarchy and role names should follow the client's existing vocabulary, because an inspection platform that uses different words from the plant is an inspection platform people argue with.

Ref	Question	Status
OQ-01	Build, buy, or configure an existing product? The reference implementation establishes that build is feasible; the decision remains commercial rather than technical.	Open — decision needed
OQ-02	Which identity provider, and is federation available for field users who have no corporate mailbox?	Open

Ref	Question	Status
OQ-03	Is WhatsApp an acceptable notification channel under the client's data protection position?	Open
OQ-04	What is the retention period for inspection evidence, and does it differ by inspection type?	Open
OQ-05	Should not-applicable answers be excluded from scoring, or counted as a pass?	Closed — excluded, with FR-FI-15 as the control
OQ-06	How many concurrent field users at peak, and across how many sites and tenants?	Open — affects sizing
OQ-07	Is geofence enforcement required, or is recording the location sufficient?	Open
OQ-08	Who owns the template library: each site, or a central quality function?	Open — affects the permission matrix
OQ-09	Is a contractor-facing external role required in the first release?	Open
OQ-10	Which asset master is authoritative, and can it be read directly?	Open
OQ-11	Is this a client-specific solution or a reusable practice asset?	Closed — client-specific
OQ-12	Pooled, schema-per-tenant or silo as the standard model? Do any early clients require contractual data separation?	Open — decision needed before section 6 is built
OQ-13	Where must each tenant's data reside, and do any tenants require a separate region?	Open
OQ-14	On what commercial and licensing basis is the service offered — subscription, fixed fee, or licence with support?	Open — drives section 15
OQ-15	Are the service targets in section 8.3 acceptable, and what service credits, if any, apply?	Open

14 Glossary

Term	Meaning
Action (CAPA)	A corrective or preventive task arising from an inspection failure or raised independently.
Break-glass access	Exceptional, time-limited access by the operator to a tenant's content, justified in writing, audited and notified to the tenant.
Critical item	A question whose failure fails the whole inspection irrespective of the numeric score.
Data Fiduciary / Data Processor	Terms from the Digital Personal Data Protection Act, 2023. The fiduciary decides why and how personal data is processed; the processor acts on its instructions.
Pass mark	The percentage a template must reach to be recorded as passed.
Pooled model	One application and database serving every tenant, with each row carrying a tenant identifier.
Recovery point / time objective	The most data that may be lost, and the longest time to restore service, after a failure.

Term	Meaning
Row-level security	A database feature that restricts which rows a connection may see, independently of the application code.
Schedule adherence	The proportion of scheduled inspections completed on or before their due date.
Segregation of duties	A control preventing one person from both performing and approving the same work.
Silo model	A dedicated application stack and database for a single tenant.
Template version	An immutable published snapshot of a template against which inspections are executed.
Tenant	One client organisation and everything that belongs to it on the platform.
Weight	The relative contribution of a scored question to the overall result.

15 Legal notices and protective provisions

Status of this section

This section is a framework of protective terms to accompany the document. It is not legal advice and does not by itself form a contract. It should be reviewed by counsel, and aligned with the signed engagement terms, before the document is issued to a third party.

Items in square brackets are to be completed. Where a signed agreement exists, its terms prevail over this section.

In this section, **the Author** means Seshail Kamanna; **FieldCheck** means the software, source code, database design, interface design, process models, documentation and demonstration materials of the platform described in this document; and **the Recipient** means any person or organisation to whom this document is made available.

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- attempt to access another tenant's data or to defeat any access control or isolation measure.

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1. This is a draft at version 0.x. Requirements are indicative until signed off in writing, and estimates, timelines and sizes are indicative only. None is a commitment, a quotation or an offer.
2. The reference implementation and any demonstration build were built to test the requirements. They are provided as they are, without any warranty of fitness for a particular purpose, availability, security or freedom from error. Warranties exist only if given in a signed agreement.
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FieldCheck is a record-keeping and workflow aid. It does not replace professional judgement, statutory or contractual inspection duties, or the Recipient's own safety management system. The Recipient remains responsible for deciding whether an item is safe, for acting on failures, for the competence of its inspectors, for checking that its templates reflect its legal and contractual obligations, and for verifying outputs before relying on them for a safety, regulatory or contractual decision.

15.6 Limitation of liability

- To the extent the law permits, the Author is not liable for indirect or consequential loss, loss of profit, revenue, business or goodwill, or loss or corruption of data, however arising.

- Subject to the next point, the Author's total liability for all claims arising from this document or any evaluation use of FieldCheck is limited to [amount or basis to be agreed — for example, fees paid for the relevant services in the preceding twelve months].
- Nothing here limits liability that cannot lawfully be limited, including liability for fraud or for death or personal injury caused by negligence.
- These limits are agreed to be reasonable. A signed agreement for a paid service may set different limits, and if so those prevail.

15.7 Data protection

This section applies when FieldCheck processes personal data, which in this platform includes names and contact details of users, photographs, signatures, and the locations at which inspections are carried out.

1. Under the Digital Personal Data Protection Act, 2023 the client organisation is the Data Fiduciary for personal data held in its tenant. The operator acts as a Data Processor, and processes that data only on the client's documented instructions and for the purpose of providing the service.
2. The operator will apply reasonable security safeguards, including those in NFR-05, NFR-06 and NFR-11, and will tell the client of a personal data breach affecting its data without undue delay and within the period the law and the agreement require [period to be agreed].
3. Sub-processors, such as the hosting provider and the email delivery service, are disclosed to the client, are bound by terms no less protective, and are changed only on notice.
4. The client chooses the region in which its tenant's data is held (OQ-13). Data is not moved outside that region except as the client instructs and the law permits.
5. The operator will assist the client in responding to requests from data principals, and will export or delete the client's data under MT-09 and MT-10, including a written confirmation of deletion.
6. Evaluation and demonstration instances contain only fictional data. Real personal data must not be loaded into them (15.3).
7. The client is responsible for having a lawful basis and any required notices and consents for the personal data it records, including photographs of people and location data.

15.8 Demonstration credentials and test data

The demonstration accounts, their shared password, and all demonstration sites, assets, users, inspections and photographs are fictional and public. They must not be used to hold real information, and must be removed or disabled before any real use. The author accepts no responsibility for data placed in an instance that still carries them.

15.9 Third-party and open-source software

FieldCheck is proprietary. It is built on open-source components, each of which remains under its own licence. The table lists the principal ones; the complete and authoritative list is the dependency lock files shipped with the build, and it should be checked before any distribution. To the Author's understanding the licences are permissive except as noted, but this is not a legal opinion.

Component	Licence
FastAPI, Pydantic, SQLAlchemy, Alembic, python-jose, React, React Router, Vite	MIT
Starlette, Uvicorn, Jinja2	BSD 3-Clause
bcrypt, python-multipart, Dexie	Apache License 2.0

Component	Licence
Pillow	HPND (permissive, MIT-style)
psycopg2	LGPL 3.0 — server builds only; not included in the Windows evaluation build
PyInstaller	GPL 2.0 with a bootloader exception — a build tool; the exception permits the executables it produces to be distributed under other terms

15.10 Reporting a security concern

A suspected vulnerability or exposure should be reported privately to [security contact address], not published. The Author will acknowledge it, investigate in good faith, and will not take action against someone who reports in good faith, accesses no more data than is needed to demonstrate the issue, and does not disclose it until it has been fixed.

15.11 Governing law and disputes

This document and any dispute about it are governed by the laws of India. The parties will first try to settle a dispute by discussion between senior representatives. Failing that, [the courts at (city), India have exclusive jurisdiction] [or: the dispute is referred to arbitration under the Arbitration and Conciliation Act, 1996, seated at (city), by a sole arbitrator] — one option to be selected in the agreement.

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